Program:

```
int x0, x1, x2, x3, x4, x5, x6, x7, x8, x9;
int delay_time = 200;

void setup()
{
  // configure pin12 as an input and enable the internal pull-up resistor
  pinMode(12, INPUT_PULLUP);

  pinMode(1, OUTPUT);
  pinMode(2, OUTPUT);
  pinMode(3, OUTPUT);
  pinMode(4, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(7, OUTPUT);
}

void loop()
{
  // read the pushbutton value into a variable
  int sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x0 = true;
```

```
}

while (x0)
{
  zero();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x1 = true;
    x0 = false;
  }
}

while (x1)
{
  one();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x2 = true;
    x1 = false;
  }
```

```
}

while (x2)
{
  two();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x3 = true;
    x2 = false;
  }
}

while (x3)
{
  three();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x4 = true;
    x3 = false;
  }
```

```
}

while (x4)
{
  four();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x5 = true;
    x4 = false;
  }
}

while (x5)
{
  five();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x6 = true;
    x5 = false;
  }
```

```
}

while (x6)
{
  six();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x7 = true;
    x6 = false;
  }
}

while (x7)
{
  seven();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x8 = true;
    x7 = false;
  }
```

```
}

while (x8)
{
  eight();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x9 = true;
    x8 = false;
  }
}

while (x9)
{
  nine();

  sensorVal = digitalRead(12);

  if (sensorVal == LOW)
  {
    x0 = true;
    x9 = false;
  }
```

```
  }
}

void zero()
{
  for (int i = 1; i < 7; i++)
  {
    digitalWrite(i, HIGH);
  }

  digitalWrite(7, LOW);
  delay(delay_time);
}

void one()
{
  digitalWrite(1, LOW);
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
  digitalWrite(6, LOW);
  digitalWrite(7, LOW);

  delay(delay_time);
}
```

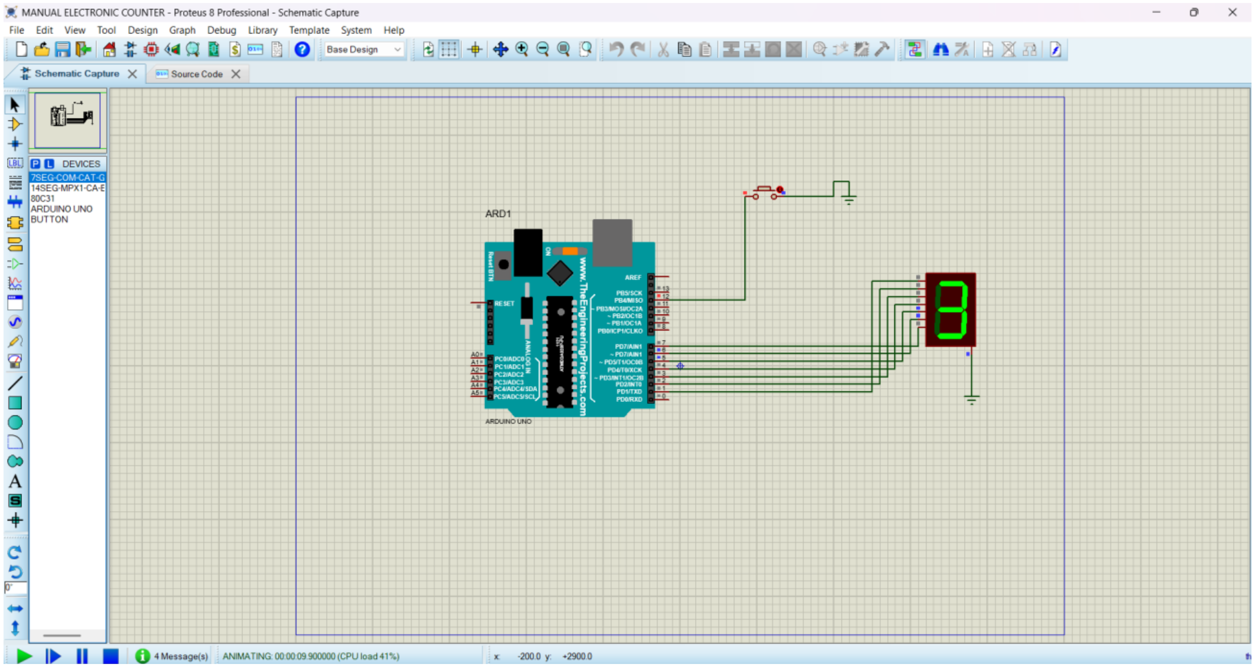
```
void two()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, HIGH);
  digitalWrite(3, LOW);
  digitalWrite(4, HIGH);
  digitalWrite(5, HIGH);
  digitalWrite(6, LOW);
  digitalWrite(7, HIGH);

  delay(delay_time);
}

void three()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, LOW);
  digitalWrite(6, LOW);
  digitalWrite(7, HIGH);

  delay(delay_time);
}
```

```
void four()
{
  digitalWrite(1, LOW);
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
  digitalWrite(6, HIGH);
  digitalWrite(7, HIGH);

  delay(delay_time);
}

void five()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, LOW);
  digitalWrite(6, HIGH);
  digitalWrite(7, HIGH);

  delay(delay_time);
}
```

```
void six()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, HIGH);
  digitalWrite(6, HIGH);
  digitalWrite(7, HIGH);

  delay(delay_time);
}

void seven()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
  digitalWrite(6, LOW);
  digitalWrite(7, LOW);

  delay(delay_time);
}
```

```
void eight()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, HIGH);
  digitalWrite(6, HIGH);
  digitalWrite(7, HIGH);
  delay(delay_time);
}

void nine()
{
  digitalWrite(1, HIGH);
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, LOW);
  digitalWrite(6, HIGH);
  digitalWrite(7, HIGH);

  delay(delay_time);
}
```

Output:


MANUAL ELECTRONIC COUNTER - Proteus 8 Professional - Schematic Capture
File
Edit
View
Tool
Design
Graph
Debug
Library
Template
System
Help
Base Design
Schematic Capture
Source Code
DEVICES
7SEG-COM-CAT-G
14SEG-MPX1-CA-E
80C31
ARDUINO UNO
BUTTON
ARD1
ARDUINO UNO
4 Message(s)
ANIMATING: 00:00:09.900000 (CPU load 41%)